53. Let XYZ be a three-digit number, where, (X+Y+Z) is not a multiple of 3. Then (XYZ + YZX + ZXY) is not divisible by

(a) 3 (b) 9
(c) 37 (d) (X + Y + Z)

Ans. (b) : Let a three-digit number = 100a + 10b + c
Which is not multiple of 3.

$\therefore$ xyz = 100a + 10b + c
yzx = 100 b + 10c + a
zxy = 100 c + 10a + b

then, (xyz + yzx + 2 xy) = 111a + 111b + 111c
(xyz + yzx + zxy) = 111 (a + b + c)
(xyz + yzx + zxy) = 3 × 37 (a + b + c)

Hence,
(xyz + yzx + zxy) is not divisible by 9.

54. Let p, q, r and s be natural numbers such that P – 2016 = q + 2017 = r – 2018 = s + 2019 Which one of the following is the largest natural number?

(a) p (b) q
(c) r (d) s

Ans. (c) : Let,
P – 2016 = q + 2017 = r – 2018 = s + 2019 = K

$\therefore$ P – 2016 = K $\Rightarrow$ P = K + 2016
q + 2017 = K $\Rightarrow$ q = K – 2017
r – 2018 = K $\Rightarrow$ r = K + 2018
s + 2019 = K $\Rightarrow$ s = K – 2019

Hence, The largest natural number = r

55. How many five-digit prime numbers can be obtained by using all the digits 1, 2, 3, 4 and 5 without repetition of digits?

(a) Zero (b) One
(c) Nine (d) Ten

Ans. (a) : Prime numbers : A prime is a number which can only be completely divided by 1 and itself.
Given digits are 1, 2, 3, 4 and 5 and we are supposed to make five digit prime numbers. We know that if the sum of the digits of a number is divisible by 3, then that number is also divisible by 3.
Here, Sum of the given digits = 1 + 2 + 3 + 4 + 5 = 15 (which is divisible by 3)
So, any number formed using these 5 digits
Will always be divisible 3.
Hence, no five-digit prime number can be made using these five-digits (without repetition)

56. A person X can complete 20% of work in 8 days and another person Y can complete 25% of the same work in 6 days. If they work together, in how many days will 40% of the work be completed?

(a) 6 (b) 8
(c) 10 (d) 12

Ans. (a) : $\because$ The taken by 'x' to do 100% work

$= \frac{8}{20} \times 100$
$= 40$ days.

Time taken by 'y' to do 100% work $= \frac{6}{25} \times 100$
$= 24$ days

According to question,

$x + y = \frac{1}{40} + \frac{1}{24}$

$(x + y) = \frac{3+5}{120}$

$(x + y) = \frac{8}{120} \Rightarrow \frac{1}{15}$ Part

Hence, The taken by '(x+y)' to do 100% work = 15 days
$\therefore$ Time taken by '(x + y)' to do 40% work

$= \frac{15}{100} \times 40$
$= \frac{15}{5} \times 2$
= 6 days.

57. A man takes half time in rowing a certain distance downstream than upstream. What is the ratio of the speed in still water to the speed of current?

(a) 1 : 2 (b) 2 : 1
(c) 1 : 3 (d) 3 : 1

Ans. (d) : Let certain distance = d
The speed of man in still water = B
and speed of current = R
According to question,

$d = (B - R)\,t$ .....(i)

and $d = (B + R)\frac{t}{2}$ .....(ii)

from eq^n (i) & (ii)

$(B - R)t = (B + R)\frac{t}{2}$
$2B - 2R = B + R$
$2B - B = R + 2R$
$B = 3R \Rightarrow \boxed{B : R = 3 : 1}$

58. How many pairs of natural numbers are there such that the difference of whose squares is 63?

(a) 3 (b) 4
(c) 5 (d) 2

Ans. (a) : Let natural numbers are x, y
then $x^2 - y^2 = 63$

$(x + y)(x - y) = 63$

for possible values

$(x + y)(x - y) = 63 \times 1$
or $(x + y)(x - y) \Rightarrow 21 \times 3$
or $(x + y)(x - y) \Rightarrow 9 \times 7$
when $(x + y)(x - y) = 63 \times 1$
$\therefore$ $x + y = 63$
$x - y = 1$
$2x = 64$
$\boxed{x = 32 \text{ and } y = 31}$

when $(x + y)(x - y) = 21 \times 3$
$x + y = 21$
$x - y = 3$
$2x = 24$
$\boxed{x = 12 \ \therefore y = 9}$

When $(x + y)(x - y) = 9 \times 7$
$x + y = 9$
$x - y = 7$
$2x = 16$
$\boxed{x = 8,\ y = 1}$

3 pairs of natural numbers are possible.

59. Which one of the following will have minimum change in its value if 5 is added to both numerator and the denominator of the fractions 2/3, 3/4, 4/5 and 5/6?

(a) 2/3 (b) 3/4
(c) 4/5 (d) 5/6

Ans. (d) : $\frac{2}{3} \Rightarrow \frac{2+5}{3+5} \Rightarrow \frac{7}{8}$

$\Rightarrow$ $\frac{7}{8} - \frac{2}{3} \Rightarrow \frac{21-16}{24} \Rightarrow \frac{5}{24} \Rightarrow 0.208$

$= \frac{3}{4} \Rightarrow \frac{3+5}{4+5} \Rightarrow \frac{8}{9}$

$\Rightarrow$ $\frac{8}{9} - \frac{3}{4} \Rightarrow \frac{32-27}{36} \Rightarrow \frac{5}{36} \Rightarrow 0.138$

$= \frac{4}{5} \Rightarrow \frac{4+5}{5+5} \Rightarrow \frac{9}{10}$

$\Rightarrow$ $\frac{9}{10} - \frac{4}{5} \Rightarrow \frac{27-24}{30} = \frac{3}{30} \Rightarrow 0.100$

$= \frac{5}{6} \Rightarrow \frac{5+5}{6+5} \Rightarrow \frac{10}{11}$

$= \frac{10}{11} - \frac{5}{6} \Rightarrow \frac{60-55}{66} \Rightarrow \frac{5}{66} \Rightarrow 0.075$

There will be minimum charge in the value obtained by doing the process in question in the fraction 5/6.

60. A digit n > 3 is divisible by 3 but not divisible by 6. Which one of the following is divisible by4?

(a) 2n (b) 3n
(c) 2n + 4 (d) 3n + 1

Ans. (d) : By option,
taking option (d)-
Required number = (3x + 1)
According to question,
A digit n > 3 is divisible by 3 but not divisible by 6.

$\therefore$ $\boxed{n = 9}$

then, (3x + 1)
Put x = 9
= (3 × 9 + 1)
= 28, Which is divisible by 4 but not divisible by 6.

61. If 1 litre of water weighs 1 kg, then how many cubic millimeters of water will weight 0.1 gm?

(a) 1 (b) 10
(c) 100 (d) 1000

Ans. (c) : $\because$ 1 litre = 1 kg
$1000\ cm^3 = 1000$ g
{$\because$ 1 litre = $1000\ cm^3$, 1 kg = 1000 g, 1 cm = 10 mm}
$1\ cm^3 = 1$g
$(10\ mm)^3 = 1$ g
$1000\ mm^3 = 1$ g
$\because$ $1\ g = 1000\ mm^3$
$\therefore$ $0.1\ g = 1000 \times 0.1\ mm^3$
$\boxed{0.1\,g = 100\,mm^3}$

62. A vessel full of water weights 40 kg. If it is one-third filled, its weight becomes 20 kg. What is the weight of the empty vessel?

(a) 10 kg (b) 15 kg
(c) 20 kg (d) 25 kg

Ans. (a) : Let the weight of vessel = x
and The weight of water= y
then $x + y = 40$ .....(i)

and $x + \frac{1}{3}y = 20$ .....(ii)

Subtracting eq^n (ii) from eq^n (i)

$y - \frac{1}{3}y = 40 - 20$

$\frac{2y}{3} = 20$

$y = 30\,kg$

Putting the value of y in eq^n (i)

$x + 30 = 40$
$\boxed{x = 10\,kg}$

Hence, the weight of vessel = 10 kg

63. A frog tries to come out of a dried well 4.5 m deep with slippery walls. Every time the frog jumps 30 cm, slides down 15 cm. what is the number of jumps required for the frog to come out of the well?

(a) 28 (b) 29
(c) 30 (d) 31

Ans. (b) : $\because$ Each Jump of frog = 30 cm
and sides every time = 15 cm
First situation,
Complete process for obtained height
= Jump – Slide
= 30 – 15
= 15 cm
Second situation,
The last complete process for obtained height
= Only Jump
Because, After reading the top of wall, the frog does not slide down.
$\therefore$ Covered height - in first situation = 450 – 30
= 420 cm
Covered height in second situation = 30 cm

$\therefore$ The total number of jumps $= \left(\frac{420}{15} + \frac{30}{30}\right)$
= 28 + 1 = 29